6. Homework Assignment

Homework Overview

- **Task**: Implement and evaluate modifications to a speech enhancement baseline using the **nanoSE** framework.
- **Dataset**: Voicebank DEMAND.
  - Clean speech mixed with various daily noise sources (office, cafe, street).
- **Deliverables**:
  - A one page report describing the experiment done and the methodology.
  - If you are in a group (2-3 students), write the names and student numbers of all members.
  - Generate the experiment report as

    ```
    python ./create_report.py runs/<baseline> runs/<myexperiment>
    ```

    and add a section describing your experiment.
  - The format can be markdown (text) or pdf.

Project Repository

Code: https://github.com/fakufaku/nanoSE

Center

Quick Start

Install Python and git. See next slide for Google Colab alternative.

```
# Get the code
git clone https://github.com/fakufaku/nanoSE.git
cd nanoSE

# Option 1: venv environment
python3 -m venv .venv
source .venv/bin/activate

# Option 2: anaconda
conda create -n nanose
conda activate nanose

# Install dependencies
pip install -r requirements.txt

# Start training
python ./train.py
```

Google Colab Alternative

If installing on your laptop is proving to be a challenge. You can try using this Google Colab.

Don't forget to enable GPU: Go to Menu > Runtime > Change runtime.

Colab has GPU!

It is FAST!

Project Ideas

The idea is to explore what is possible. Any topic is ok.

The default model is `CRNTiny` in `models/crn.py`.

1. Change the activation function in `EncoderBlock` and `DecoderBlock`.
2. Tune the step size `lr`, i.e., find the best value.
3. Find the best combination of parameter to reach maximum SNR in shortest time possible.
4. Best performance in a single epoch.
5. … (Your own idea.)

Run an Experiment

1. Make a copy of `config/default_config.py` to `config/my_experiment.py`.
2. Change things following your idea.
3. Run `python ./train.py` again.
4. This will create a new entry in `./runs`.

Generate a Report

```
python ./create_report.py --output=report/nanose.md \
  runs/experiment_a runs/experiment_b
```

This will create a report in `report/nanose.md` with a table comparing the two experiments.

1. Add your name(s) and student number(s) (team up to 3 is ok).
2. Add a section describing your experiment.
3. Add a section describing the result.

I want to try my model on real audio!

```
python enhance.py <path_to_noisy_wav> <path_to_output_wav> \
  --checkpoint runs/my_experiment/checkpoints/epoch_XXX.pt \
  --config runs/my_experiment/config.py
```

Support and Troubleshooting

1. Email me at robin.scheibler@gmail.com.
2. File an issue on nanoSE/issues

Good luck!